CLIFFT RESEARCH · JULY 2026

Two ways past the wall

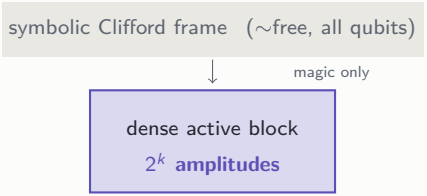
Extending clifft beyond its 2^k limit

A · Stabilizer-rank backend
change the *representation*:
pay for magic, not for dimensions

B · GPU shot batching
change the *hardware use*:
run thousands of shots abreast

clifft in thirty seconds

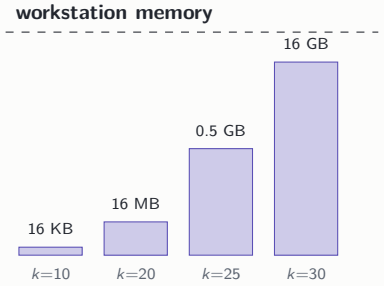
- Clifford gates are "easy": tracked **symbolically**, essentially free, at any width.
- Only *magic* (T gates) forces real amplitudes: the dense 2^k **active block**.
- k = how much magic is live at once — usually far below the qubit count.



One engine, two tiers: symbolic where physics allows it, dense only where it must be.

The wall: 2^k doubles with every step

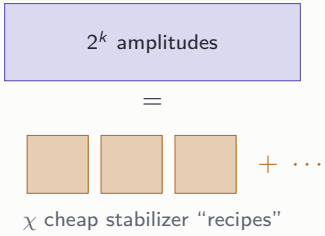
- Memory doubles per unit of k: $k = 30 \Rightarrow 16$ **GB**, $k = 40 \Rightarrow 16$ TB.
- The compiler already fights hard for small k — what still lands large is *genuinely* dense.
- So: **what do we do when 2^k itself is the cost?**



Two independent bets on what to do about it — one mathematical, one architectural.

A · The idea: pay for magic, not for dimensions

- Low-magic states are secretly simple: a **short sum of stabilizer "recipes"**, a few KB each.
- Like storing an image as a few overlapping patterns instead of every pixel.
- Cliffords are cheap on recipes; only T's multiply them; **measurements shrink them back**.



The cost driver becomes the amount of magic t, not the dimension 2^k.

A · How it fits: one compiler, two executors

- Both cost exponents are known **before anything runs** — dispatch is a one-line comparison, per program.
- **Wired, not just proposed**: the backend runs clifft's optimized circuits; exact clifft stays the default, the backend is an explicit opt-in with an error dial δ.

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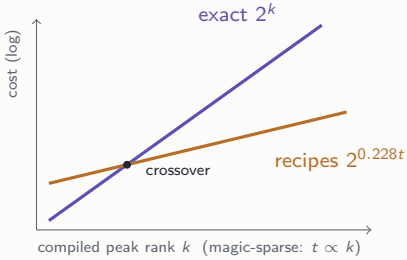
clifft compiles; the cheaper engine runs — on the same optimized circuit.

A · Why it should work

- An optimal T-split keeps the recipe count at $2^{0.228\,t}$ — far slower growth than 2^k.
- A **tunable** error δ lets you keep only a random sample. Measured: error $= 0.50\,\delta$, on the nose.
- Same language clifft already speaks: frame + measurement collapse.



Trade exactness you can measure for reach you cannot otherwise have.

A · It generalizes: any Clifford+T circuit

- Circuits with magic *everywhere*? Teleport each in-line T onto a helper qubit — all magic becomes **one up-front layer** again, with **no penalty for interleaving**.
- Tested on circuits whose answer is known *by construction* (hidden shift): up to **96 qubits**, right answer in seconds.
- Often clifft still wins such circuits outright — and the dispatch rule *knows in advance*.

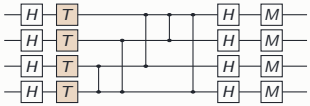
Generality at the product-magic price — validated end to end against exact references.

A · Recycled magic: pay per episode, not per program

- Long computations **recycle** magic: the block grows, is measured out, grows again — "episodes". Total magic grows with runtime; **per-episode magic stays bounded**.
- Run one episode at a time: pay $2^{\text{per-episode}}$, not 2^{total}, with **zero extra error**. Measured: **72 qubits at ≤ 400 recipes** where whole-circuit needs 2^{72}.
- Compressing further also works (measured, honest limits) — the dispatcher prices it.

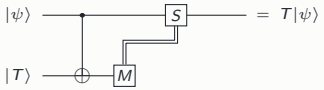
The circuit's rhythm — magic in, magic measured out — is itself a computational resource.

A · The circuits behind the numbers



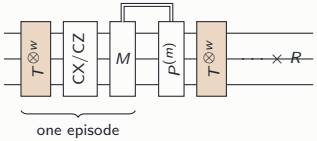
Dense IQP — the stress test

magic on every qubit; *compiles* to peak rank $n/2$ (crossover, frontier)



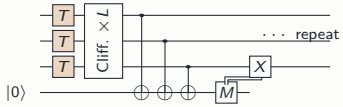
T-gadget — the generality trick

in-line $T \rightarrow$ ancilla + measurement + conditional S (hidden shift, 96 qubits)



Magic conveyor — recycled magic

w T's in, measured out with feedforward, R rounds (episodic)



Cultivation patch — adaptive

syndrome rounds + logical Clifford traffic while magic accumulates (the frame's home turf)

A stress test, a trick, a rhythm, a protocol — each drives one benchmark.

A · Evidence (measured, against clifft's real fast path)

- Everything validated to **machine precision**; benchmarked against clifft's *real* fast path.
- Crossover at $n \approx 34$–48 (coarse–8% error); past $n \approx 62$ the **only method that runs at all**.
- External tools: QuiZX 25× slower by $n{=}56$; Quokka# **times out beyond** $n{=}24$.

$n = 72$	memory	time
any exact method	∼1 TB	—
rank backend	**1.9 GB**	**267 s**

Accuracy dial, measured:
error $= 0.50\,\delta$ across the whole sweep
(halve the error ⇒ 4× the work).

The niche is real — and honestly bounded: below the crossover, exact clifft stays the right tool.

A · Where it wins, where it doesn't

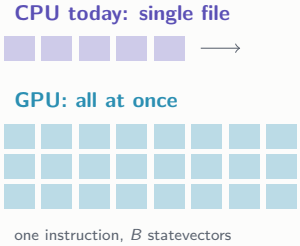
Small peak rank	Large peak rank, sparse magic	Adaptive, Clifford-heavy
clifft is faster *and* exact — keep it. (Most circuits, however wide, compile here.)	**The niche**: the only feasible method, accuracy dial-controlled.	**Real protocols win** 2–16× — and the standard open-source tool cannot run them at all.

- Also measured where it *loses* (up to ∼2× overhead) — the advantage is a measured boundary, not a slogan.

Status: prototype validated; honest benchmarks, natural workloads, external baselines done.

B · The idea: don't run one shot faster — run 4,096 abreast

- One shot's small array **cannot keep a GPU busy**: a stadium choir asked to sing one sentence.
- But workloads are **thousands of shots** of the same program — today run one after another.
- Put B=4,096 shots side by side; every gate hits *all of them* in one launch.



Batching doesn't make the GPU faster — it makes the problem big enough for the GPU to be fast.

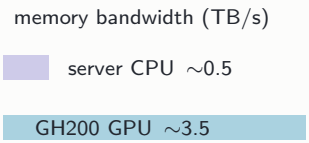
B · Why it should work (1): shots march in lockstep

- A shot's shape — what runs when, how big the array is — is **identical for every shot**, fixed at compile time by clifft's compiler.
- Per-shot randomness is just a few numbers: **SIMD lanes, one per shot**. The classic "shots diverge" objection *does not apply here*.
- And nobody serves this niche: existing batched simulators **fall back to one-at-a-time** on mid-circuit measurement — clifft's home turf.

The hardest architectural question resolves in the design's favor before any GPU code is written.

B · Why it should work (2): the speed limit is bandwidth — and GPUs have 7× more

- These kernels do almost no math per byte: **memory speed is the speed limit**, on any machine.
- Measured: 11 CPU cores = only 2.6× one core — the bus, not the cores.
- A GH200 has ∼7× the bandwidth ⇒ **5–10× more shots/s**, honestly bounded.



big published "100×" numbers use weak baselines — we keep a refuted-claims list and quote the physics.

The claim is deliberately modest — which is why we can trust it.

B · The one risk, and the $20 experiment that settles it

- Mid-circuit measurement needs the CPU: each crossing costs microseconds — **the same scale as the kernels themselves**.
- But it is paid **once per batch** (nanoseconds per shot), on a link built to make it cheap.
- A faithful microbenchmark — *measurements and round-trips included* — is built, validated, and waiting. **GH200 time: $2/hour.**

round-trip measured cheap	**go**: batched backend; first target = replay-style queries (no round-trips at all)
round-trip eats the win	**no-go**, cleanly — or on-device sampling is the follow-up

Either answer is cheap, quantitative, and final — that is what the microbenchmark is for.

How the two bets fit together

A · Stabilizer rank extends *reach*

circuits clifft cannot hold at all today —
approximate, with a measured accuracy dial

B · GPU batching extends *throughput*

the circuits clifft already runs —
many more shots per second, exact as ever

- Both work *because of* clifft's structure: the frame, the small block, the compile-time schedule.
- Both were reviewed adversarially; every headline number survived a fair baseline.

Next: A — benchmarks complete; write it up. B — the GH200 run, then batched replay queries.